Name: _____] Roll No: _____

INTRODUCTION TO DATA SCIENCE

Instructor: Dr M. Nadeem Majeed

SECTION 1: MCQs (MULTIPLE CHOICE QUESTIONS)

Total Marks: 10

Q1. A developer accidentally pushed a commit containing sensitive configuration data to main. To safely undo this in a shared remote repository without rewriting history, which command should be used?

A) git reset --hard HEAD~1
B) git revert <commit-hash>
C) git checkout HEAD~1
D) git stash pop

Q2. You are halfway through implementing a feature when an urgent production bug arises. You run 'git stash' to clean your workspace. Where does Git store your uncommitted changes?

A) In the local repository commit tree as a permanent commit
B) In the remote repository branch
C) In a temporary stash storage stack managed by Git
D) In the staging index area

Q3. While inspecting an older release, a teammate executes 'git checkout <commit-hash>'. What state is their local repository currently in?

A) Staging state
B) Detached HEAD state
C) Merge conflict state
D) Rebase lock state

Q4. In a CI/CD workflow, why might a developer run 'git fetch' and inspect the changes before integrating them?

A) git fetch automatically resolves merge conflicts
B) git fetch updates remote-tracking references without integrating changes into the current branch
C) git fetch deletes untracked local files
D) git fetch automatically creates a release tag

Q5. You modified 'style.css' in your working directory, but the changes are incorrect and you want to discard the uncommitted changes in that file, restoring it to the current commit. Which command should you use?

A) git restore style.css
B) git rm style.css
C) git commit --amend
D) git status --clean style.css

Q6. What happens if you access a nonexistent key in a Python dictionary using dict[key]?

A) It returns None
B) It returns 0
C) It raises a KeyError
D) It automatically inserts the key with a value of None

Q7. You need to store a fixed collection of settings (host, port, database_name) that should not be modified after creation. Which Python data structure is most appropriate?

A) List
B) Tuple
C) Dictionary
D) Set

Q8. Which built-in Python function allows you to iterate through a sequence while keeping track of both the index and the item?

A) range(len(items))
B) enumerate(items)
C) zip(items)
D) map(items)

Q9. In Python, which block is designed for cleanup code that should execute whether or not an exception occurs?

A) try
B) except
C) else
D) finally

Q10. What is the output of the expression s[::-1] if s = "DataScience"?

A) "DataScience"
B) "ecneicSataD"

C) "D"
D) IndexError: string index out of range

SECTION 2: GITHUB PRACTICAL ASSESSMENT

Total Marks: 10 | Total Questions: 5 | Marks per Question: 2

Q1:You are working on a team project. Your teammate has pushed new commits to the remote GitHub repository, but you want to inspect those changes before integrating them into your local branch.

Write the Git command you would use to download the latest remote updates without integrating them into your current branch. Then explain the difference between 'git fetch' and 'git pull', including what happens to your current branch when each command is executed.

Q2:During a feature deployment, you run 'git merge feature/task-style' into 'main', and Git reports a merge conflict in 'README.md'. Write the step-by-step procedure to inspect the conflict markers, resolve the conflict, and finalize the merge commit.

Q3:You created a local commit with the message "Temp changes", but immediately realized you forgot to stage an important file. Explain how 'git reset --soft HEAD~1' allows you to undo the commit while retaining the changes in the staging area, and contrast it with 'git reset --hard'.

Q4:You have been assigned a GitHub Issue to implement a new feature in the Student Task Manager app. Describe the standard collaborative workflow, including creating a

feature branch, committing and pushing changes, opening a Pull Request (PR), obtaining code review and approval, and using an automatic issue-closing keyword.

Q5:Your team has finished testing the Student Task Manager app and is ready to mark a stable production milestone. Write the Git commands to create an annotated tag named 'v1.0.0' for the current commit and push the tag to GitHub. Explain how this tag can be used when creating a GitHub Release.

SECTION 3: PYTHON PRACTICAL ASSESSMENT

Case Study: Student Cafeteria & Expense System

Total Marks: 10 | Total Questions: 5 | Marks per Question: 2

A university cafeteria tracks student transactions and monthly allowances using Python data structures. Each student record is a dictionary containing the student's name, monthly allowance (in Rs.), a list of individual order expense amounts (in Rs.), and their scholarship status.

Dataset:

```
students = [
{"name": "Ali", "allowance": 10000, "expenses": [1500, 2200, 1800, 3000], "category": "Regular"},
{"name": "Sara", "allowance": 8000, "expenses": [2500, 3100, 2800], "category": "Scholarship"},
{"name": "Usman", "allowance": 12000, "expenses": [1000, 1200], "category": "Regular"},
{"name": "Ayesha", "allowance": 6000, "expenses": [2000, 2500, 1800], "category": "Scholarship"},
{"name": "Zain", "allowance": 15000, "expenses": [4000, 5000, 4500], "category":
```

"Regular"}
]

Q1:Write Python code using a loop to calculate and print each student's name, total expenditure (sum of the items in their 'expenses' list), and remaining balance (allowance minus total expenditure).

Q2:Write Python code to iterate through 'students' and display the message "WARNING: High Expenditure!" for any student whose total expenditure exceeds 80% of their monthly allowance.

Q3:Scholarship students receive a 10% cashback bonus on their remaining allowance balance. Write Python code to filter all "Scholarship" category students and construct a new dictionary named 'scholarship_bonuses', where each key is the student's name and each value is 10% of their remaining balance.

Q4:Student "Sara" makes a new cafeteria purchase of Rs. 500. Write Python code using dictionary lookup and list operations to locate Sara's record in 'students' and append 500 to her 'expenses' list.

Q5:Write Python code using a loop and conditions to find and display the name of the student who has spent the highest total amount across the dataset, along with their calculated total expenditure.